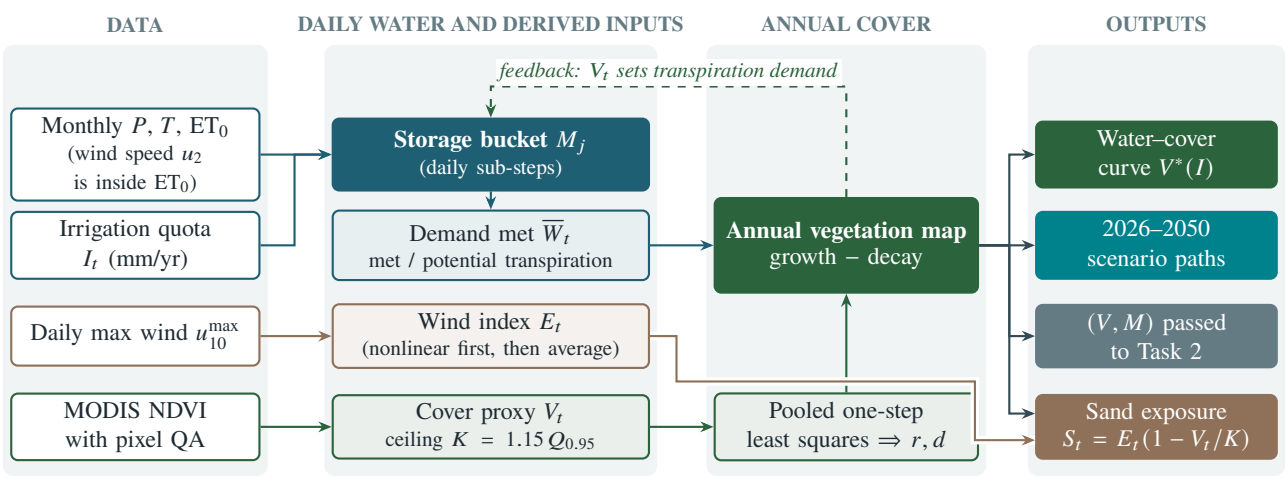

DATA
DAILY WATER AND DERIVED INPUTS
ANNUAL COVER
OUTPUTS
feedback: V_t sets transpiration demand
Monthly P, T, ET_0 (wind speed u_2 is inside ET_0)
Irrigation quota I_t (mm/yr)
Daily max wind u_{10}^{max}
MODIS NDVI with pixel QA
Storage bucket M_j (daily sub-steps)
Demand met $\overline{W}_t$ met / potential transpiration
Wind index E_t (nonlinear first, then average)
Cover proxy V_t ceiling $K = 1.15\,Q_{0.95}$
Annual vegetation map growth − decay
Pooled one-step least squares $\Rightarrow r, d$
Water–cover curve $V^*(I)$
2026–2050 scenario paths
(V, M) passed to Task 2
Sand exposure $S_t = E_t(1 - V_t/K)$